



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2016 Preliminary Course Yearly Examination

31st of August, 2016

General instructions

- Working time – 2 hours 30 minutes (plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.

SECTION I

- Questions 1 - 8
- Mark your answers in the section provided in your answer booklet

SECTION II

- Questions 9 - 16
- Commence each new question on a **new page**. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: # BOOKLETS USED:

Class (please ✓)

☐ Mr Berry

☐ Miss Lee

☐ Mr Hwang

☐ Mr Lin

☐ Mr Ireland

☐ Ms Ziaziaris

☐ Dr Jomaa

Marker's use only.

QUESTION	1–8	9	10	11	12	13	14	15	16	Total
MARKS	$\overline{8}$	$\overline{8}$	$\overline{11}$	$\overline{9}$	$\overline{18}$	$\overline{14}$	$\overline{17}$	$\overline{15}$	$\overline{15}$	$\overline{115}$

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Section I: Objective response

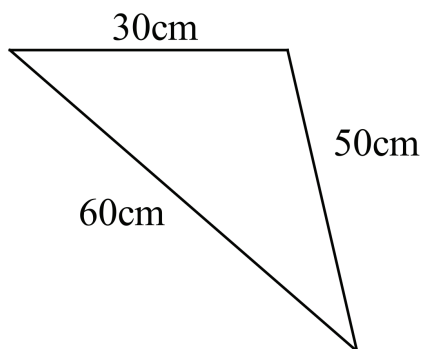
Mark your answers on the multiple choice section in the answer booklet.

Marks

1. What is the value of $\frac{(1.49)^2 - 1.98}{\sqrt{11.62 + 8.34 \times 2.72}}$ correct to three significant figures? **1**
- (A) 0.40
- (B) 0.041
- (C) 0.0409
- (D) 0.0410
2. What is the centre and radius of the circle with equation $x^2 + y^2 + 6x - 8y - 11 = 0$? **1**
- (A) Centre $(-3, -4)$ and radius 36
- (B) Centre $(-3, 4)$ and radius 36
- (C) Centre $(-3, -4)$ and radius 6
- (D) Centre $(-3, 4)$ and radius 6
3. What are the coordinates of the focus of the parabola $x^2 = 2(y - 1)$? **1**
- (A) $(0, \frac{1}{2})$
- (B) $(0, \frac{3}{2})$
- (C) $(\frac{1}{2}, 0)$
- (D) $(\frac{3}{2}, 0)$
4. What is the value of $\frac{a}{b}$ if the lines $ax + 2y = 6$ and $4y = bx - 9$ are parallel? **1**
- (A) $\frac{1}{2}$
- (B) $-\frac{1}{2}$
- (C) -2
- (D) 2

5. The following triangle has sides 30cm, 50cm and 60 cm.

1



Angle C is the largest angle. Which of the following expressions is correct for angle C ?

- (A) $\cos C = \frac{30^2 + 60^2 - 50^2}{2 \times 30 \times 60}$
- (B) $\cos C = \frac{50^2 + 30^2 - 60^2}{2 \times 50 \times 30}$
- (C) $\cos C = \frac{50^2 + 60^2 - 30^2}{2 \times 50 \times 60}$
- (D) $\cos C = \frac{50^2 + 30^2 - 60^2}{2 \times 50 \times 60}$
6. What is the solution to the equation $\cos(\frac{\theta}{2} + 20^\circ) = \sin \theta$ for $0^\circ \leq \theta \leq 90^\circ$?

1

- (A) $\theta = 20^\circ$
- (B) $\theta = 40^\circ$
- (C) $\theta = \frac{160^\circ}{3}$
- (D) $\theta = \frac{140^\circ}{3}$

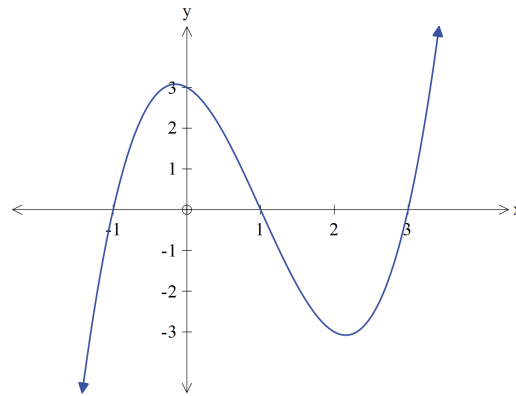
7. What is the minimum value of $\cos \theta + 2 \cos(\theta + 240^\circ)$?

1

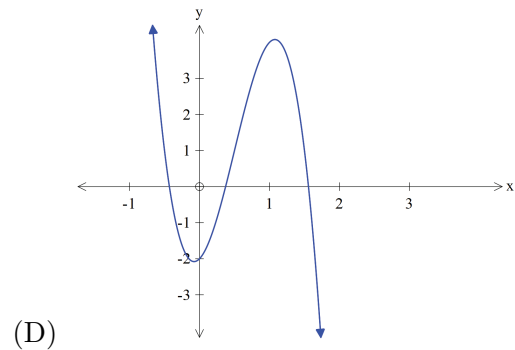
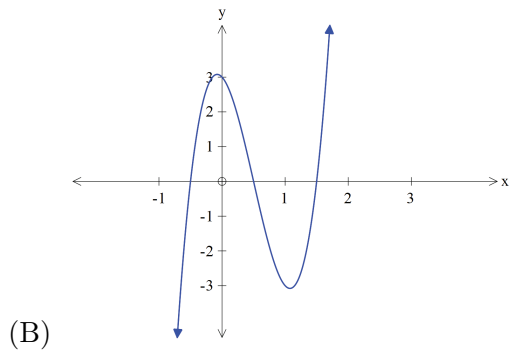
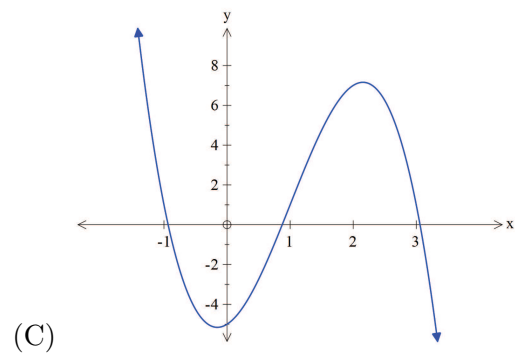
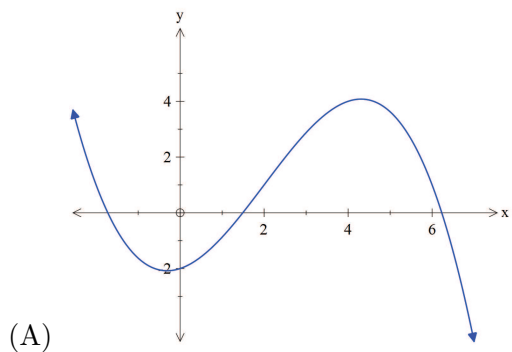
- (A) $-2\sqrt{3}$
- (B) $-\sqrt{3}$
- (C) $-\frac{\sqrt{3}}{2}$
- (D) $-\frac{\sqrt{3}}{4}$

8. The graph of a function f , with domain R , is as shown.

1



The graph which best represents $1 - f(2x)$ is



End of Section I.
Examination continues overleaf.

Section II

Write your answers in the writing booklet supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (8 Marks) Commence a NEW page. **Marks**

- (a) Show that $\frac{4}{2 + \sqrt{5}} - \frac{1}{9 - 4\sqrt{5}}$ is rational **3**
- (b) Solve $|2x - 1| = 3x$ **3**
- (c) Solve $|5 - 3x| \geq 4$ **2**

Question 10 (11 Marks) Commence a NEW page. **Marks**

- (a) If $\sin \theta = \frac{3}{4}$ and $90^\circ \leq \theta \leq 180^\circ$ find the exact value of $\sec \theta$ **2**
- (b) i. Show $\frac{\cos x + \sin x}{\cos x - \sin x} + \frac{\cos x - \sin x}{\cos x + \sin x} = \frac{2}{2 \cos^2 x - 1}$ **3**
- ii. Use the above result to solve **3**
- $$\frac{\cos x + \sin x}{\cos x - \sin x} + \frac{\cos x - \sin x}{\cos x + \sin x} = 4 \text{ for } 0^\circ \leq \theta \leq 360^\circ$$
- (c) Sketch the region defined by $(x - 2)^2 + y^2 < 4$ and $y \geq (x - 2)^2$ **3**
(i.e. Find the intersection of these two graphs)

Question 11 (9 Marks) Commence a NEW page. **Marks**

- (a) Sketch the following showing essential features
- i. $y = \frac{3}{x + 1}$ **2**
- ii. $y = |x - 2| + 4$ **2**
- (b) i. Find the domain of $y = \sqrt{1 - x^2}$ **1**
- ii. Find the domain of $y = \sqrt{x^2 - 1}$ **1**
- iii. Find the range of $y = \sqrt{x^2 - 1}$ **1**
- (c) State whether $y = \frac{x^3}{1 + x}$ is an odd function, an even function or neither, **2**
justifying your answer with working out.

Question 12 (18 Marks)

Commence a NEW page.

Marks

- (a) Differentiate
- i. $y = (2 - 3x)^5$ **1**
 - ii. $y = \frac{3x}{\sqrt{x}}$ **2**
 - iii. $y = \frac{x^3 - 2x}{x^2}$ **2**
 - iv. $y = x(x - 3)^7$ **2**
- (b) Find the equation of the normal to the curve $y = 2x^2 - 4$ at the point where $x = 1$. **3**
- (c) Given the function $y = x^3 - 3x^2 - 12$
- i. Find the stationary points and determine their nature. **4**
 - ii. Find any points of inflexion. **2**
 - iii. Hence, sketch the curve. **2**

Question 13 (14 Marks)

Commence a NEW page.

Marks

- (a) Solve the inequality $\frac{x}{2x - 1} \leq 5$ **3**
- (b) Find the acute angle between the lines $2x - 3y + 1 = 0$ and $y = 2x - 1$ to the nearest degree **2**
- (c) The point $P(5, 7)$ divides the interval AB joining the points $A(-1, 1)$ and $B(3, 5)$ in the ratio $m : 1$. Find the value of m **2**
- (d) Without using calculus, sketch on a number plane the graph of $y = \frac{x - 1}{x^2 - 9}$. Label any asymptotes and the x and y intercepts. **3**
- (e) Find the domain of the function $f(x) = \sqrt{4 - x^2} + \frac{1}{x + 2}$ **2**
- (f) If $\sin 2A = \frac{1}{2}$, find the exact value of $\frac{1}{\sin A \cos A}$ **2**

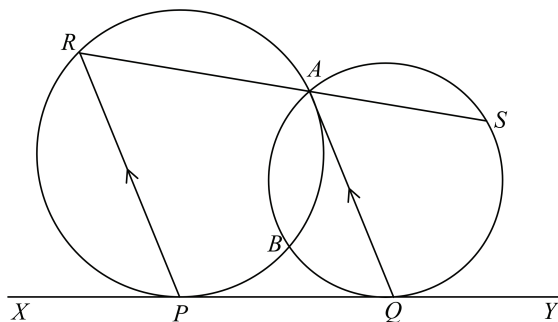
Question 14 (17 Marks)	Commence a NEW page.	Marks
(a) Find the equation of the straight line passing through $A(3, -2)$ inclined at an angle of 120° with the positive direction of the x -axis. (Leave values in exact form)		2
(b) Solve for x the equation $4^x - 3(2^x) + 2 = 0$		3
(c) Find the values of k for which $kx^2 + (k + 3)x + 4$ is positive definite		3
(d) One of the roots of the equation $2x^2 - 15x + c = 0$ is four times the other. Find the roots of the equation and hence the value of c		3
(e) i. Prove that $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$		2
ii. If the roots of the quadratic $x^2 + 4x + 2 = 0$ are α and β , using the result in part (i) or otherwise, find a value for	$\frac{1}{\alpha^3} + \frac{1}{\beta^3}$	2
(f) Find the Cartesian equation of a curve whose parametric equations are $x = 2 \cos \theta$ and $y = \sqrt{3} \sin \theta$. (You do not need to simplify).		2

Question 15 (15 Marks)

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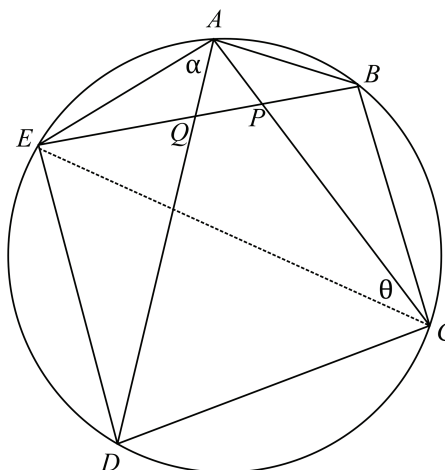
Marks

- (a) The point $A(2at, at^2)$ lies on the parabola with equation $x^2 = 4ay$
- Find the equation of the tangent to the parabola at A **2**
 - Find the coordinates of B and C , the x and y -intercepts of this tangent. **2**
 - Show that B is the midpoint of AC **2**
- (b) Two circles meet in A and B . PQ is a common tangent and $PR \parallel QA$. RA is produced to meet the other circle in S .

**DIAGRAM NOT TO SCALE**

Copy the diagram into your writing booklet.

- Prove that $PRSQ$ is cyclic **2**
 - Prove that $PA \parallel QS$ **2**
- (c) $ABCDE$ is a pentagon inscribed in a circle, where $AB = AE$, BE meets AC and AD at P and Q respectively.

**DIAGRAM NOT TO SCALE**Given $\angle ACE = \theta$ and $\angle DAE = \alpha$,

- Copy the diagram onto your answer booklet and show that $\angle AEB = \theta$, giving reasons **2**
- Hence, or otherwise, show that $CPQD$ is a cyclic quadrilateral **3**

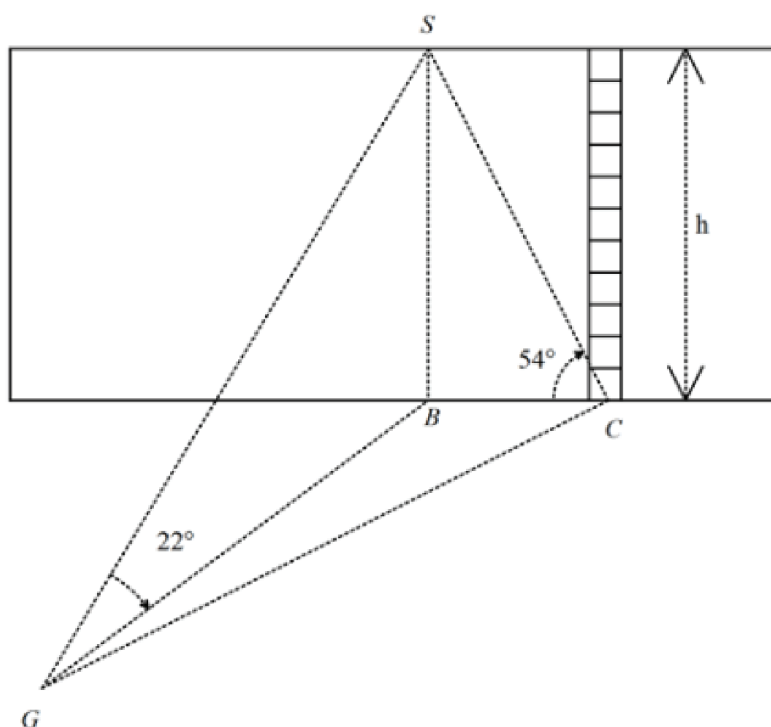
Question 16 (15 Marks)

Commence a NEW page.

Marks

- (a) i. Expand $\tan(A + B)$ **1**
 ii. The roots of $x^2 - 2x - 1 = 0$ are $\tan A$ and $\tan B$. If $A + B$ is acute, find the size of $A + B$ **3**

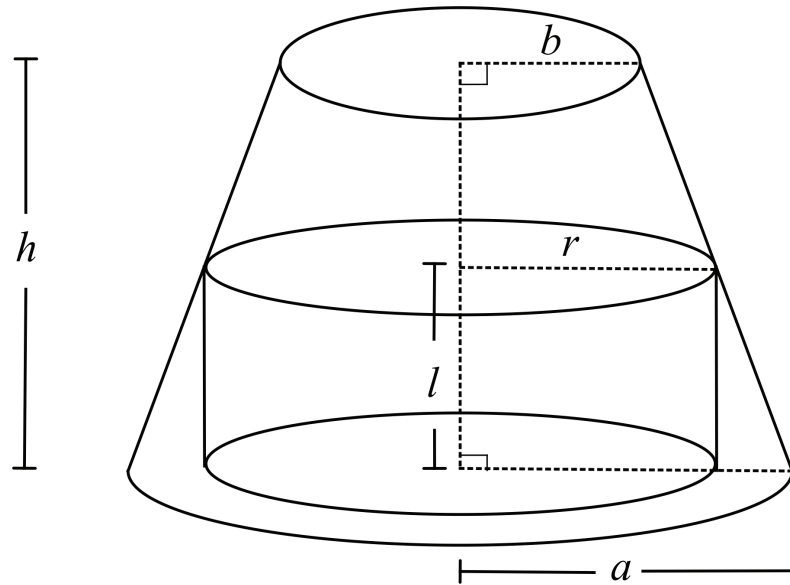
- (b) A child climbs down a ladder and into a gorilla enclosure at Cincinnati zoo. A gorilla (point G on the diagram) notices the child (point C on the diagram) standing at the bottom of the ladder. A zookeeper (point S on the diagram) is standing on top of the enclosure wall. The zookeeper aims a tranquiliser at the gorilla with an angle of depression of 22° and will shoot the gorilla if he approaches the child. The angle of elevation from the child to the zookeeper is 54° . The distance of the gorilla from the child is 3 times the height of the wall. The wall runs East-West.



- i. Show that $GB = h \cot 22^\circ$ **2**
 ii. Similarly, write an expression for the length of BC . **1**
 iii. Hence, or otherwise, find the bearing of the child from the gorilla **3**

- (c) A solid block of wood is in the shape of a frustrum of a right circular cone, the radii of the end faces being a and b ($a > b$). The frustrum has fixed height h . A right circular cylinder of height l is to be cut from the frustrum, with its axis along the axis of the cone.

5



Show that the volume of the cylinder is a maximum when the height of the cylinder is

$$\frac{ah}{3(a-b)}$$

End of paper.

2016 Ext 1 Prelim Yearly - Suggested Solutions

MCQ

1. D
2. D
3. B
4. B
5. B
6. D
7. B
8. D

Question 9

$$\begin{aligned} 3 \quad a) & \frac{4(2-\sqrt{5})}{4-5} - \frac{(9+4\sqrt{5})}{81-16\times 5} \\ &= \frac{8-4\sqrt{5}}{-1} - \frac{(9+4\sqrt{5})}{1} \\ &= 4\sqrt{5}-8-9-4\sqrt{5} \\ &= -17. \end{aligned}$$

$$\begin{aligned} 3 \quad b) & |2x-1| = 3x \\ & 2x-1 = 3x \quad \text{OR} \quad 2x-1 = -3x \\ & \cancel{2x}-1 = x \quad \text{OR} \quad 5x = 1 \\ & \quad \quad \quad x = \frac{1}{5} \end{aligned}$$

Check $x = -1$

$$\begin{aligned} |2(-1)-1| &= -3 \\ &\text{not poss.} \end{aligned}$$

Check $x = \frac{1}{5}$

$$|2 \times \frac{1}{5} - 1| = 3(\frac{1}{5})$$

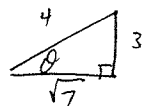
Only soln $x = \frac{1}{5}$.

$$\begin{aligned} 2 \quad c) & |5-3x| \geq 4 \\ & 5-3x \geq 4 \quad \text{OR} \quad 5-3x \leq -4 \\ & -3x \geq -1 \quad \quad -3x \leq -9 \\ & x \leq \frac{1}{3} \quad \text{OR} \quad x \geq 3. \end{aligned}$$

Question 10

2

a) $\sin \theta = \frac{3}{4}$



$\sec \theta = -\frac{4}{\sqrt{7}}$

3

b) (i) LHS =
$$\frac{\cos x + \sin x}{\cos x - \sin x} + \frac{\cos x - \sin x}{\cos x + \sin x}$$
$$= \frac{(\cos x + \sin x)^2 + (\cos x - \sin x)^2}{\cos^2 x - \sin^2 x}$$
$$= \frac{\cos^2 x + 2\sin x \cos x + \sin^2 x + \cos^2 x - 2\sin x \cos x + \sin^2 x}{\cos^2 x - \sin^2 x}$$
$$= \frac{2}{\cos^2 x - \sin^2 x}$$
$$= \frac{2}{\cos^2 x - (1 - \cos^2 x)}$$
$$= \frac{2}{2\cos^2 x - 1}$$
$$= \text{RHS.}$$

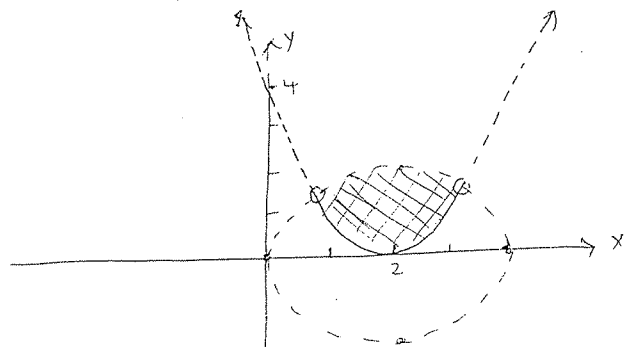
3

(ii) $\therefore \frac{2}{2\cos^2 x - 1} = 4$
$$2 = 8\cos^2 x - 4$$
$$8\cos^2 x = 6$$
$$\cos^2 x = \frac{3}{4}$$
$$\cos x = \pm \frac{\sqrt{3}}{2}$$

$\therefore x$ lies in all quads
 $x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

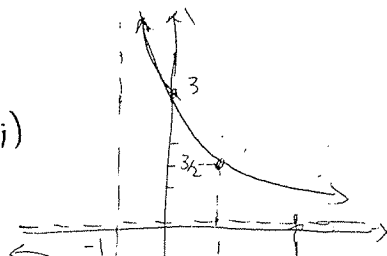
3

c)

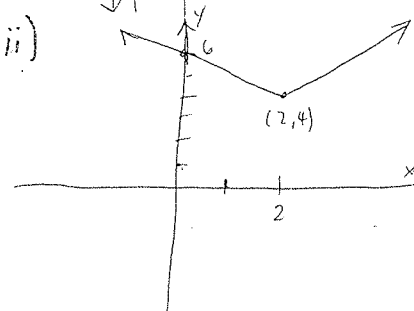


Question 11

2 a) i)



2 ii)



Question 11 continued

1 b) i) $y = \sqrt{1-x^2}$
 $D: -1 \leq x \leq 1$

1 ii) $y = \sqrt{x^2-1}$
 $x^2-1 \geq 0$
 $(x-1)(x+1) \geq 0$

 $x \leq -1, x \geq 1$

1 iii) $y \geq 0$

2 c) $y = \frac{x^3}{1+x}$
 $f(a) = \frac{a^3}{1+a}$
 $f(-a) = \frac{(-a)^3}{1+(-a)}$
 $= \frac{-a^3}{1-a}$
 $\therefore f(a) = \frac{-a^3}{1+a}$

Neither.

Question 12

1 a) (i) $5(2-3x)^4 - 3$
 $= -15(2-3x)^4$

2 (ii) $\frac{3x}{\sqrt{x}}$
 $= 3x^{\frac{1}{2}}$
 $y' = \frac{1}{2} \times 3 x^{-1/2}$
 $= \frac{3}{2} x^{-1/2}$
 $= \frac{3}{2\sqrt{x}}$

2 (iii) $\frac{x^3}{x^2} - \frac{2x}{x^2}$
 $x - \frac{2}{x}$
 $x - 2x^{-1}$
 $= 1 + 2x^{-2}$
 $= 1 + \frac{2}{x^2}$

2 (iv) $(x-3)^7 \cdot 1 + x \cdot 7(x-3)^6$
 $= (x-3)^7 + 7x(x-3)^6$
 $(x-3)^6(8x-3)$

3 b) $y = 2x^2 - 4$
 $\frac{dy}{dx} = 4x$

At $x=1, y=-2$

$\frac{dy}{dx} = 4$

So m of normal $= -\frac{1}{4}$

Eqn of normal

$y+2 = -\frac{1}{4}(x-1)$

$4y+8 = -x+1$

4

$$c) i) y = x^3 - 3x^2 - 12$$

$$\frac{dy}{dx} = 3x^2 - 6x$$

$$\frac{d^2y}{dx^2} = 6x - 6$$

Stat Pts: $\frac{dy}{dx} = 0$

$$3x^2 - 6x = 0$$

$$3x(x-2) = 0$$

$$x=0, \quad x=2$$

$$y=-12, \quad y=-16$$

Test (0, -12)

$$\frac{d^2y}{dx^2} = -6$$

$$\frac{d^2y}{dx^2} < 0 \therefore \boxed{\text{max}(0, -12)}$$

Test (2, -16)

$$\frac{d^2y}{dx^2} = 6$$

$$\frac{d^2y}{dx^2} > 0 \therefore \boxed{\text{min}(2, -16)}$$

2 ii) Inflex. Pts: $\frac{d^2y}{dx^2} = 0$ and change in sign.

$$6x - 6 = 0$$

$$x = 1$$

$$y = -14$$

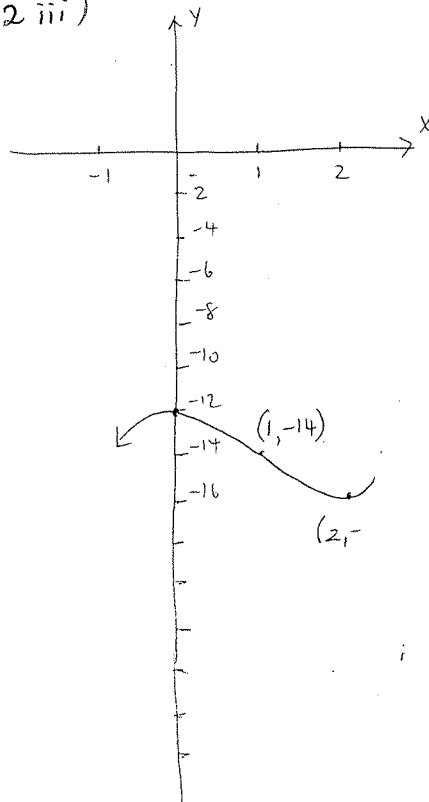
Test (1, -14)

x	0	1	2
$\frac{d^2y}{dx^2}$	-6	0	6

$\therefore$ Inflex Pt at (1, -14)

Y-int: $x=0$
 $y=-12$

2 iii)



Question 13

3 a)

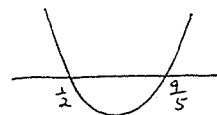
$$\frac{x}{2x-1} \leq 5 \quad x \neq \frac{1}{2}$$

$$x(2x-1) \leq 5(2x-1)^2$$

$$0 \leq (2x-1)[5(2x-1) - x]$$

$$0 \leq (2x-1)(10x-5-x)$$

$$0 \leq (2x-1)(9x-5)$$



$$\therefore x < \frac{1}{2} \text{ or } x \geq \frac{5}{9}$$

2 b)

$$m_2 = \frac{2}{3}, \quad m_1 = 2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - \frac{2}{3}}{1 + 2(\frac{2}{3})} \right|$$

$$= \frac{4}{7}$$

$$\therefore \theta = 30^\circ$$

c)

$$A(-1, 1) \quad B(3, 5)$$

$$m = 1$$

$$\frac{-1 + 3m}{m+1} = 5 \quad (1)$$

$$\frac{1 + 5m}{m+1} = 7 \quad (2)$$

$$-1 + 3m = 5m + 5$$

$$-6 = 2m$$

$$\therefore m = -3$$

$$1 + 5m = 7m + 7$$

$$2m$$

$$3 \text{ d)} \quad y = \frac{x-1}{x^2-9}$$

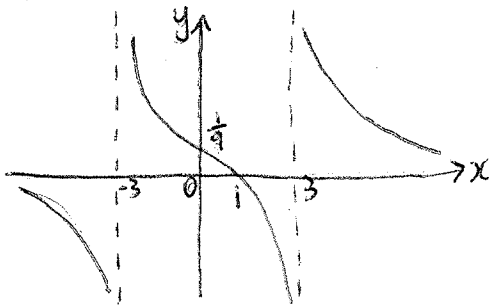
$$= \frac{x-1}{(x+3)(x-3)}$$

Vertical asymptotes: $x = \pm 3$

Horizontal asymptotes: $y = 0$

x-intercept: $(1, 0)$

y-intercept: $(0, \frac{1}{9})$



3 e) Domain of $\sqrt{4-x^2}$ is $-2 \leq x \leq 2$
Domain of $\frac{1}{x+2}$ is $x \neq -2$

$\therefore$ domain of $f(x)$ is $-2 < x \leq 2$

$$2 \text{ f)} \quad \frac{1}{\sin A \cos A} = \frac{2}{\sin 2A}$$

$$= 2 \div \frac{1}{2}$$

$$= 4$$

Question 14 continued.

$$2 \text{ e) i)} \quad RHS = (a+b)^3 - 3ab(a+b)$$

$$= (a+b)[(a+b)^2 - 3ab]$$

$$= (a+b)[a^2 + 2ab + b^2 - 3ab]$$

$$= (a+b)(a^2 - ab + b^2)$$

$$= a^3 + b^3$$

$$= LHS$$

$$2 \text{ ii)} \quad \frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{\beta^3 + \alpha^3}{\alpha^3 \beta^3}$$

$$= \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{(\alpha\beta)^3}$$

$$= \frac{(-4)^3 - 3(2)(-4)}{2^3}$$

$$= -5$$

$$2 \text{ f)} \quad x = 2 \cos \theta \rightarrow \cos \theta = \frac{x}{2}$$

$$y = \sqrt{3} \sin \theta \rightarrow \sin \theta = \frac{y}{\sqrt{3}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{y}{\sqrt{3}}\right)^2 + \left(\frac{x}{2}\right)^2 = 1$$

Question 14

2 a) $m = \tan \theta$
 $= \tan 120^\circ$
 $= -\sqrt{3}$

$$y + 2 = -\sqrt{3}(x - 3)$$

3 b) $4^x - 3(2^x) + 2 = 0$

Let $2^x = u$

$$u^2 - 3u + 2 = 0$$

$$(u-2)(u-1) = 0$$

$$u = 1 \quad u = 2$$

$$2^x = 1 \quad 2^x = 2$$

$$\therefore x = 0, 1$$

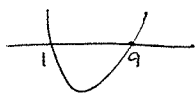
2 c) Positive definite: coeff. of $x^2 > 0$ and $\Delta < 0$

i.e. $k < 0$ and $(k+3)^2 - 4(k)(4) < 0$

$$k^2 + 6k + 9 - 16k < 0$$

$$k^2 - 10k + 9 < 0$$

$$(k-9)(k-1) < 0$$



$$\therefore 1 < k < 9$$

3 d) Let the roots be $\alpha, 4\alpha$

$$\alpha + 4\alpha = \frac{15}{2}$$

$$10\alpha = 15$$

$$\alpha = \frac{3}{2}$$

$$4\alpha = 6$$

$$4\alpha^2 = \frac{c}{2}$$

$$9 = \frac{c}{2}$$

$$c = 18$$

Question 15

2 a) i) $y = \frac{x^2}{4a}$
 $\frac{dy}{dx} = \frac{2x}{4a}$
 $= \frac{x}{2a}$

When $x = 2at$, $m = \frac{2at}{2a}$
 $= t$

Equation of tangent:

$$y - y_1 = m(x - x_1)$$

$$y - at^2 = t(x - 2at)$$

$$= tx - 2at^2$$

$$\therefore \boxed{y = tx - at^2}$$

2 ii) $B(at, 0)$
 $C(0, -at^2)$

Midpoint AC:

$$x = \frac{2at + 0}{2}$$

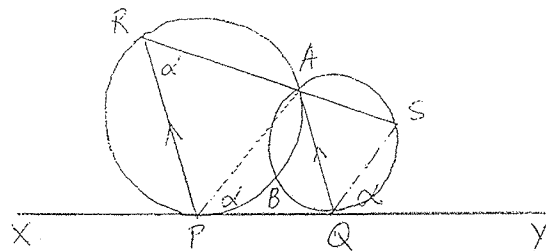
$$= at$$

$$y = \frac{at^2 - at^2}{2}$$

$$= 0$$

$\therefore B$ is the midpoint of AC

Question 15(b)



(i) Join SQ

Let $\angle SQY = \alpha$

then $\angle QAS = \alpha$ (angle in alternate segment)
 $\angle ARP = \alpha$ (corresponding angles in parallel lines $AQ \parallel RP$)

$\therefore \angle SQY = \angle ARP$

$\therefore PQSR$ is a cyclic quad (exterior angle of cyclic quad equals opposite interior angle).

(ii) Join PA

$\angle APQ = \angle ARP$ (angle in alternate segment)

$\therefore \angle APQ = \angle SQY$ (both α from (i).)

$\therefore AP \parallel SQ$ (corresponding angles are equal)

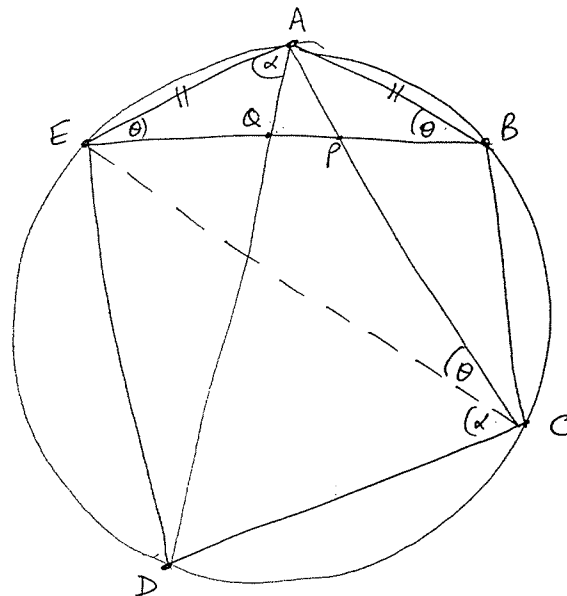
Question 15 continued

C)

2 i) $\hat{ACE} = \hat{ABE} = \theta$ (Angles in same segment) ✓
 $\hat{ABE} = \hat{AEB}$ (Equal L's opp. equal sides, $AE = AB$)
 $\therefore \hat{AEB} = \theta$ (given)

3 ii) $\hat{DAE} = \hat{DCE} = \alpha$ (Angles in same segment)
 $\hat{AQB} = \hat{DAE} + \hat{AEB}$ (Exterior L' of $\triangle ADE$)
 $= \alpha + \theta$ (equal to sum of two opp. interior L's).

As $\hat{PCD} = \hat{AQB} = \alpha + \theta$, $PCDQ$ is a cyclic quad. as the exterior angle is equal to the interior opposite angle.



Question 16

1 a) i) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

3 ii) $\tan A + \tan B = 2$
 $\tan A \tan B = -1$
 $\tan(A+B) = \frac{2}{1 - (-1)}$
 $= 1$
 $\therefore A+B = 45^\circ$

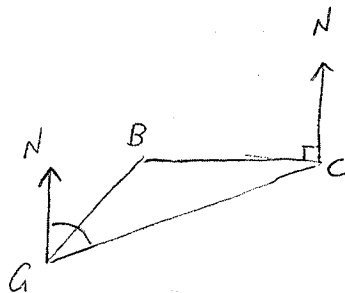
b) i) $\tan 22 = \frac{h}{GB}$
 $GB = \frac{h}{\tan 22}$
 $= h \cot 22$

ii) $BC = h \cot 54$

iii) $\cos \hat{B}CG = \frac{BC^2 + GC^2 - GB^2}{2(BC)(GC)}$
 $= \frac{(h \cot 54)^2 + (3h)^2 - (h \cot 22)^2}{2(h \cot 54)(3h)}$
 $= \frac{\cot^2 54 + 9 - \cot^2 22}{6 \cot 54}$

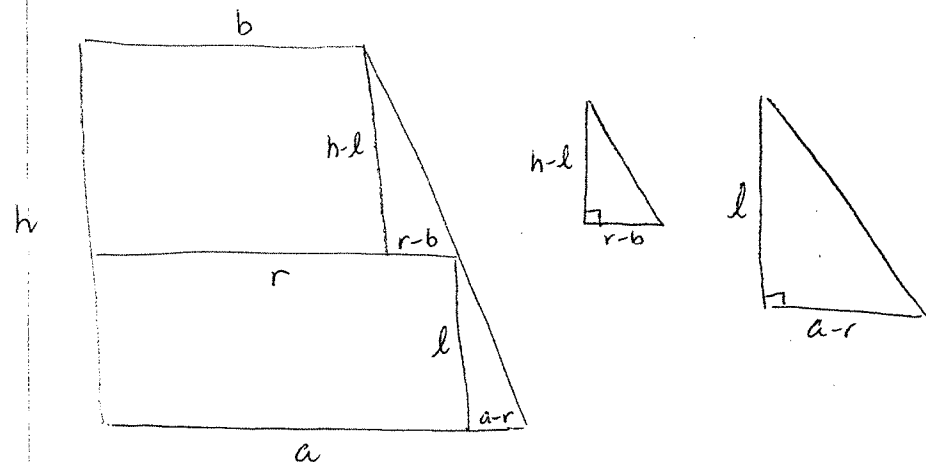
$\hat{B}CG = \cos^{-1} \left(\frac{\cot^2 54 + 9 - \cot^2 22}{6 \cot 54} \right)$
 $= 38^\circ 42' \text{ (to nearest min.)}$

$\therefore$ bearing is $180^\circ - 90^\circ - 38^\circ 42' = 51^\circ 18' \text{ (to nearest min.)}$



Method 1

5 c) $a, h, r, (a-b) > 0$



By similar triangles,

$$\frac{h-l}{l} = \frac{r-b}{a-r}$$

$$(h-l)(a-r) = l(r-b)$$

$$ah - hr - al + rl = rl - bl$$

$$ah - hr = al - bl$$

$$\therefore l = \frac{h(a-r)}{(a-b)}$$

The volume of the cylinder is:

$$V = \pi r^2 l$$

$$= \pi r^2 h \frac{(a-r)}{a-b}$$

$$= \frac{\pi h}{a-b} [r^2(a-r)]$$

$$= \frac{\pi h}{a-b} [ar^2 - r^3]$$

$$\frac{dV}{dr} = \frac{\pi h}{a-b} [2ar - 3r^2]$$

$$\frac{d^2V}{dr^2} = \frac{\pi h}{a-b} [2a - 6r]$$

$$\frac{dV}{dr} = 0$$

$$2ar - 3r^2 = 0$$

$$r(2a - 3r) = 0$$

$$r \neq 0, \quad r = \frac{2a}{3}$$

$$\begin{aligned} \text{When } r = \frac{2a}{3}, \quad V'' &= \frac{\pi h}{a-b} \left[2a - 6\left(\frac{2a}{3}\right) \right] \\ &= \frac{\pi h}{a-b} [-4a] \end{aligned}$$

$$< 0 \text{ since } a, h, r, (a-b) > 0$$

$\therefore$ Volume is maximum when $r = \frac{2a}{3}$.

$$\text{and } l = \frac{h}{a-b} \left(a - \frac{2a}{3} \right)$$

$$= \frac{h}{a-b} \left(\frac{a}{3} \right)$$

$$= \frac{ah}{3(a-b)}$$

Method 2

$$r = \frac{ah - al + bl}{h}$$

$$V = \pi \left(\frac{ah - al + bl}{h} \right)^2$$

$$\begin{aligned} \frac{dV}{dl} &= \pi \left(\frac{ah - al + bl}{h} \right)^2 + 2\pi l \left(\frac{b-a}{h} \right) \left(\frac{ah - al + bl}{h} \right) \\ &= \pi \left(\frac{ah - al + bl}{h} \right) \left[\frac{ah - al + bl + 2l(b-a)}{h} \right] \\ &= \pi \left[\frac{ah + (b-a)l}{h} \right] \left[\frac{ah + 3l(b-a)}{h} \right] \end{aligned}$$

$$\frac{dV}{dl} = 0 \therefore h = \frac{ah}{a-b} \text{ or } l = \frac{ah}{3(a-b)}$$

$$\frac{d^2V}{dl^2} = \frac{\pi(b-a)}{h^2} [4ah + 6l(b-a)]$$

$$\text{at } l = \frac{ah}{3(a-b)}, \quad \frac{d^2V}{dl^2} = \frac{\pi(b-a)}{h^2} [4ah - 2ah] < 0$$

since $b-a < 0$

$$l = \frac{ah}{a-b}, \quad \frac{d^2V}{dl^2} = \frac{\pi(b-a)}{h^2} [4ah - 6ah] > 0$$

$$\therefore l = \frac{ah}{3(a-b)}$$

Method 3

Let r denote the radius of the right circular cylinder and l its height.

Then the volume is given by

$$V = \pi r^2 l$$

l lies on the line connecting the origin to the top of the frustrum. The equation of this line is $y = \frac{h}{a-b} x$ since it has slope $\frac{h}{a-b}$

(since the height of the frustrum is h and the horizontal distance between the lower edge & upper edge is $a-b$). Therefore

$$l = \frac{h(a-r)}{a-b}$$

Hence the volume of the cylinder is given by

$$V = \pi r^2 \left(\frac{h(a-r)}{a-b} \right)$$

$$= \frac{\pi r^2 (ha - hr)}{a-b}$$

$$= \frac{\pi r^2 ha - \pi hr^3}{a-b}$$

$$\frac{dV}{dr} = \frac{2\pi r ha - 3\pi hr^2}{a-b}$$

$$\frac{dV}{dr} = 0 : \quad 2\pi r ha - 3\pi hr^2 = 0$$

$$\pi rh(2a - 3r) = 0$$

$$r = 0 \quad \text{or} \quad r = \frac{2a}{3}$$

$$\frac{d^2V}{dr^2} = \frac{2\pi ha - 6\pi hr}{a-b} < 0 \quad \text{when} \quad r = \frac{2a}{3}$$